

Quine McCluskey

- The Quine McCluskey, also called tabulation method, is a very useful and convenient tool for simplification of Boolean functions for large numbers of variables.
- As we know that the [Karnaugh map method](#) is a very useful and convenient tool for simplification of Boolean functions as long as the number of variables does not exceed four.
- But for case of large number of variables, the visualization and selection of patterns of adjacent cells in the Karnaugh map becomes complicated and too much difficult.
- For those cases Quine McCluskey tabulation method takes vital role to simplify the Boolean expression.
- The Quine McCluskey tabulation method is a specific step-by-step procedure to achieve guaranteed, simplified standard form of expression for a function.

Prime Implicant and Essential Prime Implicant Terms

- **Implicant** is defined as a group of ones, and
- **Prime implicant** is the largest possible group of ones.
- **Essential prime implicant** is the prime implicant having at least one minterm which cannot be combined in any other way.
 - This means that the essential prime implicant is represented in minimized form, which cannot be combined further.

Example

$$Y(A, B, C, D) = \sum m(0,1,3,7,8,9,11,15)$$

Step 1: Group minterms **based on the number of ones** in their binary equivalent representation.

Group	Minterm	Binary Representation(ABCD)
Group 0	M0	0000
Group 1	M1, M8	0001 1000
Group 2	M3, M9	0011 1001
Group 3	M7, M11	0111 1011
Group 4	M15	1111

Step 2. Form Matched Pairs (two numbers differing in only one bit) for all groups.

- For example, M0 and M1, (0000-0001), are matched pair because they differ in only one digit.
- The rule for selecting a group is n and $n+1$ group. This means that group 1 is checked for matched pair for group 0, the group 2 for group 1, and so on.

Group	Matched Pairs	Binary Representation(ABCD)
Group 0	M0-M1	000*
	M0-M8	*000
Group 1	M1-M3	00*1
	M1-M9	*001
	M8-M9	100*
Group 2	M3-M7	0*11
	M3-M11	*011
	M9-M11	10*1
Group 3	M7-M15	*111
	M11-M15	1*11

Step 3. Find matched pairs out of the matched pairs found in previous step.

- Group (M0-M1) with (M8-M9), the representation is (000*) and (100*) first bit is different whereas rest three bits are the same. So they are qualified as matched pairs.

Group	Matched Pairs	Binary Representation(ABCD)
Group 0	M0-M1-M8-M9	*00*
	M0-M8-M1-M9	*00*
Group 1	M1-M3-M9-M11	*0*1
	M1-M9-M3-M11	*0*1
Group 2	M3-M7-M11-M15	**11
	M3-M11-M7-M15	**11

Step 4. Again the same step is repeated to find out the matched pair between group 0 and group 1 as obtained in step 3.

- We can see that no matched pairs exists between the groups 0,1, and 2.
Therefore we can now write the minimized form.
- From group 0, we can see that the variables A and D are eliminated. Only B and C are left. If it is 0 represent it with a compliment otherwise represent as normal variable.
- So the prime implicant is obtained in the last column.

Group	Matched Pairs	Binary Representation(ABCD)	Minimized Form
Group 0	M0-M1-M8-M9 M0-M8-M1-M9	*00* *00*	$\bar{B}\bar{C}$
Group 1	M1-M3-M9-M11 M1-M9-M3-M11	*0*1 *0*1	$\bar{B}D$
Group 2	M3-M7-M11-M15	**11 **11	CD

Step 5. Obtain the essential prime implicants out of the prime implicants.

- From the prime implicants, identify the minterms involved. E.g. for the first one the minterms involved are (0,1,8,9),
- Put a cross if it appears in the minterm given in the problem and put a circle if that minterm is appearing only once.
- The minterm involved in the second row (1,3,9,11) are not essential prime implicants. Since each of them has already appeared in the first or third row.

		Minterms given in the problem							
Prime Implicant	Minterm Involved	0	1	3	7	8	9	11	15
$\bar{B}\bar{C}$	(0,1,8,9)	⊗	X			⊗	X		
$\bar{B}D$	(1,3,9,11)		X	X			X	X	
CD	(3,7,11,15)			X	⊗			X	⊗

So finally the minimized expression can be given as

$$Y = \bar{B}\bar{C} + CD$$

Example 2:

$$F = \sum m(0, 1, 2, 3, 5, 7, 8, 10, 14, 15)$$

Decimal	Binary
0	0000
1	0001
2	0010
3	0011
5	0101
7	0111
8	1000
10	1010
14	1110
15	1111

<u>Number</u>	<u>Number of 1s</u>	<u>Binary Number</u>
1 2 8	1	0001 0010 1000
3 5 10	2	0011 0101 1010
7 14	3	0111 1110
15	4	1111

<u>Number</u>	<u>Number of 1s</u>	<u>Binary Number</u>	1st column	
1	1	0001	(0,1)	000-
2		0010	(0,2)	00-0
8		1000	(0,8)	-000
3	2	0011	1,3	00-1
5		0101	1,5	0-01
10		1010	2,3	001-
			2,10	-010
			8,10	10-0
7	3	0111	3,7 5,7 10-14	0-11
14		1110		01-1
				1-10
15	4	1111	7,15 14,15	-111 111-

<u>Number</u>	<u>Number of 1s</u>	<u>Binary Number</u>	1st column		2nd column	
1	1	0001	(0,1)	000-	0,1,2,3	00--
2		0010	(0,2)	00-0	0,2,1,3	00--
8		1000	(0,8)	-000	0,8,2,10	-0-0
3	2	0011	1,3	00-1	0,2,8,10	-0-0
5		0101	1,5	0-01	1,3,5,7	0--1
10		1010	2,3	001-		
7	3	0111	2,10	-010		
14			8,10	10-0		
			3,7	0-11		
	4	1111	5,7	01-1		
			10-14	1-10		
15			7,15	-111		
			14,15	111-		

<u>Number</u>	<u>Number of 1s</u>	<u>Binary Number</u>	<u>1st column</u>		<u>2nd column</u>	
1	1	0001	(0,1)	000-	0,1,2,3	00--
2		0010	(0,2)	00-0	0,2,1,3	00--
8		1000	(0,8)	-000	0,8,2,10	-0-0
3	2	0011	1,3	00-1	0,2,8,10	-0-0
5		0101	1,5	0-01	1,3,5,7	0--1
10		1010	2,3	001-		
7	3	0111	2,10	-010		
14		1110	8,10	10-0		
15		1111	3,7	0-11		
			5,7	01-1		
			10-14	1-10		
			7,15	-111		
			14,15	111-		

[illegible]

Combination	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0,1,2,3	x	x	x	x												
0,8,2,10	x		x						x		x					
1,3,5,7		x		x		x		x								
10,14											x				x	
7,15								x								x
14,15															x	x

$$0,8,2,10 \quad (-0-0) = \overline{B}\overline{D}$$

$$1,3,5,7 \quad (0--1) = \overline{A}D$$

$$14,15 \quad (111-) = ABC$$

$$F = \overline{B}\overline{D} + \overline{A}D + ABC$$